HOW MIGHT WE MAKE UNIVERSITY COURSES MORE ACCESSIBLE FOR STUDENTS WITH DISABILITIES IN CANADA?

DESIGN PROCESS



Research



Define



Empathize



Storyboarding

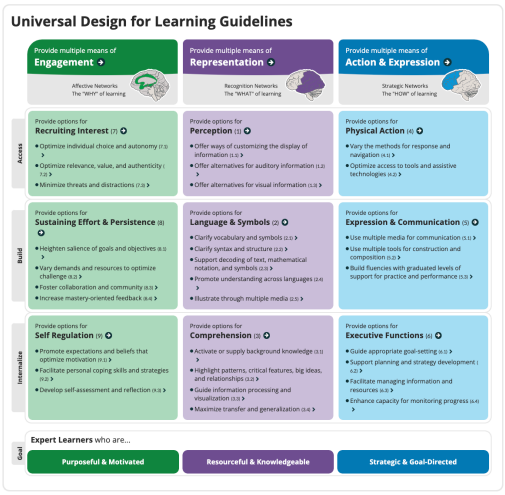


Brainstorming

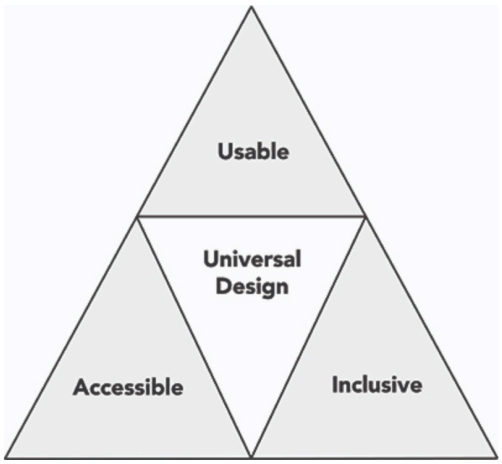


Low-Fidelity Prototype

RESEARCH



https://udlguidelines.cast.org



https://www.washington.edu/doit/equal-access-universal-design-instruction

SUBJECT MATTER EXPERT INTERVIEW:

Janet Arnold

Academic Advisor and Professor at Mount Royal University

Your chances of knowing someone with learning disabilities are very good. Did you know?



- 2.3 million students are diagnosed with specific learning disabilities (SLD) and receive services under IDEA. This represents 35% of all students receiving special education services.*
- 75% – 80% of special education students identified as LD have their basic deficits in language and reading***
- 60% of adults with severe literacy problems have undetected or untreated learning disabilities**

https://ldaamerica.org/support/new-to-ld/

UNIVERSITY OF TORONTO

https://teaching.utoronto.ca/teaching-support/udl/

MOUNT ROYAL UNIVERSITY 1910

http://mtroyal.ca/AcademicSupport/ResourcesServices/AccessibilityServices/UniversalDesign.htm

UNIVERSITY OF CALGARY

https://taylorinstitute.ucalgary.ca/resources/universal-design-learning-higher-education#:~:text=Universal Design for Learning is,they are engaged in learning.

Academic Articles

Dalton, E. M., Lyner-Cleophas, M., Ferguson, B. T., & Mckenzie, J. (2019). Inclusion, universal design and universal design for learning in higher education: South Africa and the United States. *African Journal of Disability*, 8. doi:10.4102/ajod.v8i0.519

Hromalik, C. D., Myhill, W. N., Ohrazda, C. A., Carr, N. R., & Zumbuhl, S. A. (2021). Increasing Universal Design for Learning knowledge and application at a community college: The Universal Design for Learning Academy. *International Journal of Inclusive Education*, 1-16. doi:10.1080/13603116.2021.1931719

Mcmahon, D. D., & Walker, Z. (2019). Leveraging Emerging Technology to Design an Inclusive Future with Universal Design for Learning. *Center for Educational Policy Studies Journal*, 9(3), 75-93. doi:10.26529/cepsj.639



DEFINE

Design Brief

Problem Statement How might we make university courses more accessible for students with disabilities in Canada?	
Design Statement Design a poster, pamphlet or presentation with information on how to increase universal design for learning practices in university courses in Canada	**Who Will Benefit** • People with disabilities (both diagnosed and undiagnosed) • University professors
Considerations This prototype would be shown to university professors and would be written in plain language to improve accessibility. Should be optimized for visual impairments (Ex: alt text).	**Key Partnerships** • University instructors • Subject-matter experts ○ Accessibility Services ○ Non-profit organizations
Opportunities This prototype could be presented at conventions, professor staff meetings, conferences and non-profit events	**Key Activities** • Storyboarding ○ User personas • Framing the design brief • Critique session • Prototyping
Constraints The design must provide tangible solutions for professors to implement, lack of funding available to do more of an intricate design such as an app or website, and time limited to under 2 weeks	**Key Resources** • IDEO mindsets & methods • Classmates and Instructor • Subject-matter experts • Community resources

3 EMPATHIZE

User Personas



Name: Onyx

Profession: Student

Age: 19

Onyx is starting their university degree at a Canadian university.

They struggled throughout their high school years, but still managed to get good enough grades to get into a Bachelor of Arts program, with a major in Philosophy



Name: Dr. Harris

Profession: Philosophy Professor

Age: 56

Dr. Harris is a professor in philosophy at a Canadian university. He has taught for over 20 years and has not changed his teaching method much since he started teaching, using Powerpoint slides for lessons. He evaluates his students using an essay, midterm and final exam

4 STORYBOARDING

1



Onyx is starting their university degree and is excited to start their learning journey at a Canadian university

2



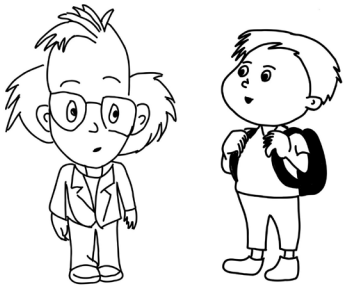
In Onyx's first lecture, they meet their professor, Dr. Harris, who teaches using conventional teaching methods

3



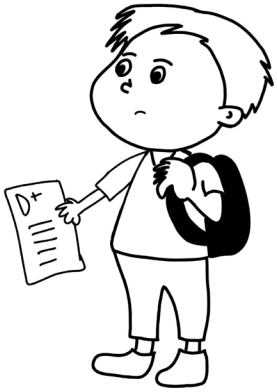
Onyx quickly feels overwhelmed and starts to question whether they made the right decision by going to university

4



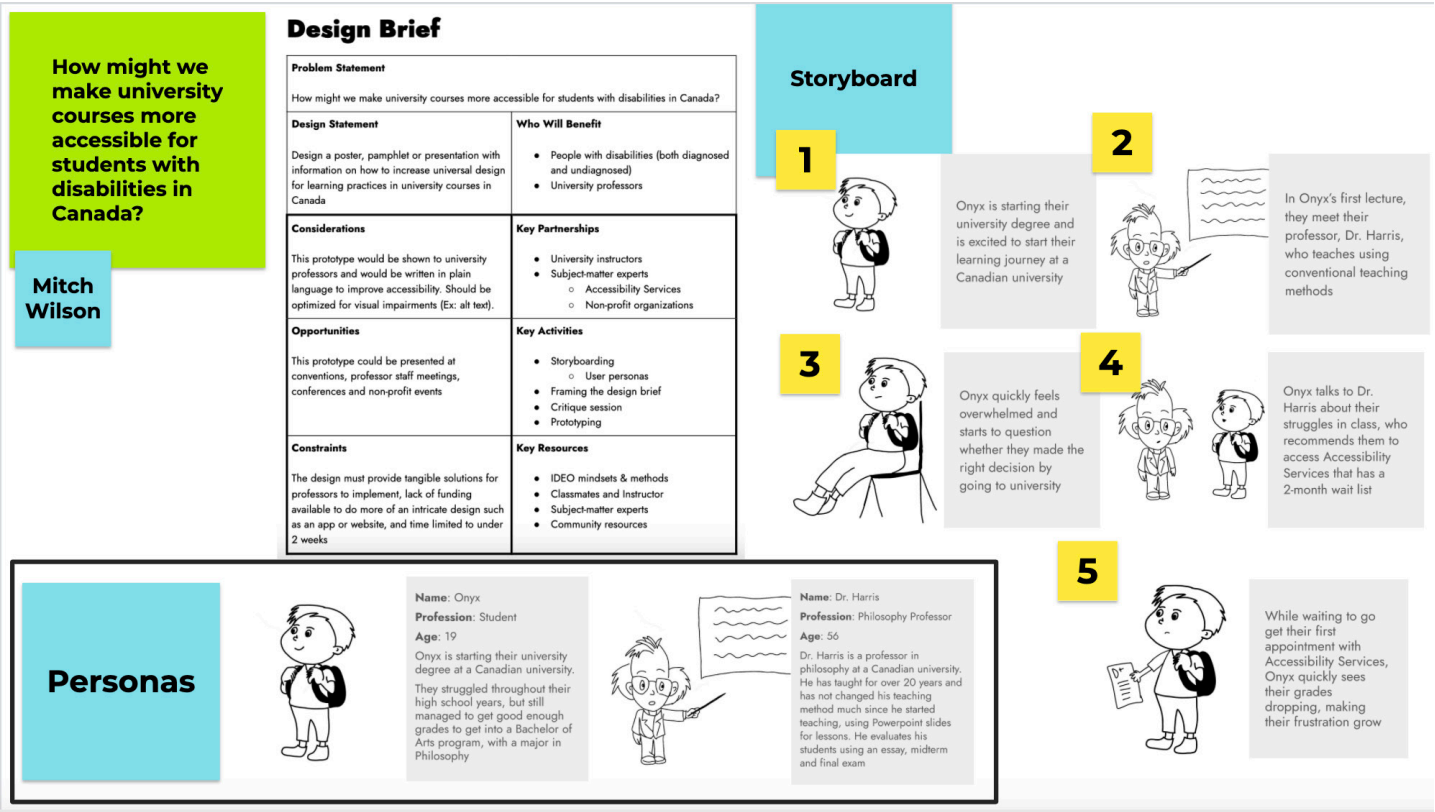
Onyx talks to Dr. Harris about their struggles in class, who recommends them to access Accessibility Services that has a 2-month wait list

5



While waiting to go get their first appointment with Accessibility Services, Onyx quickly sees their grades dropping, making their frustration grow

5 BRAINSTORMING



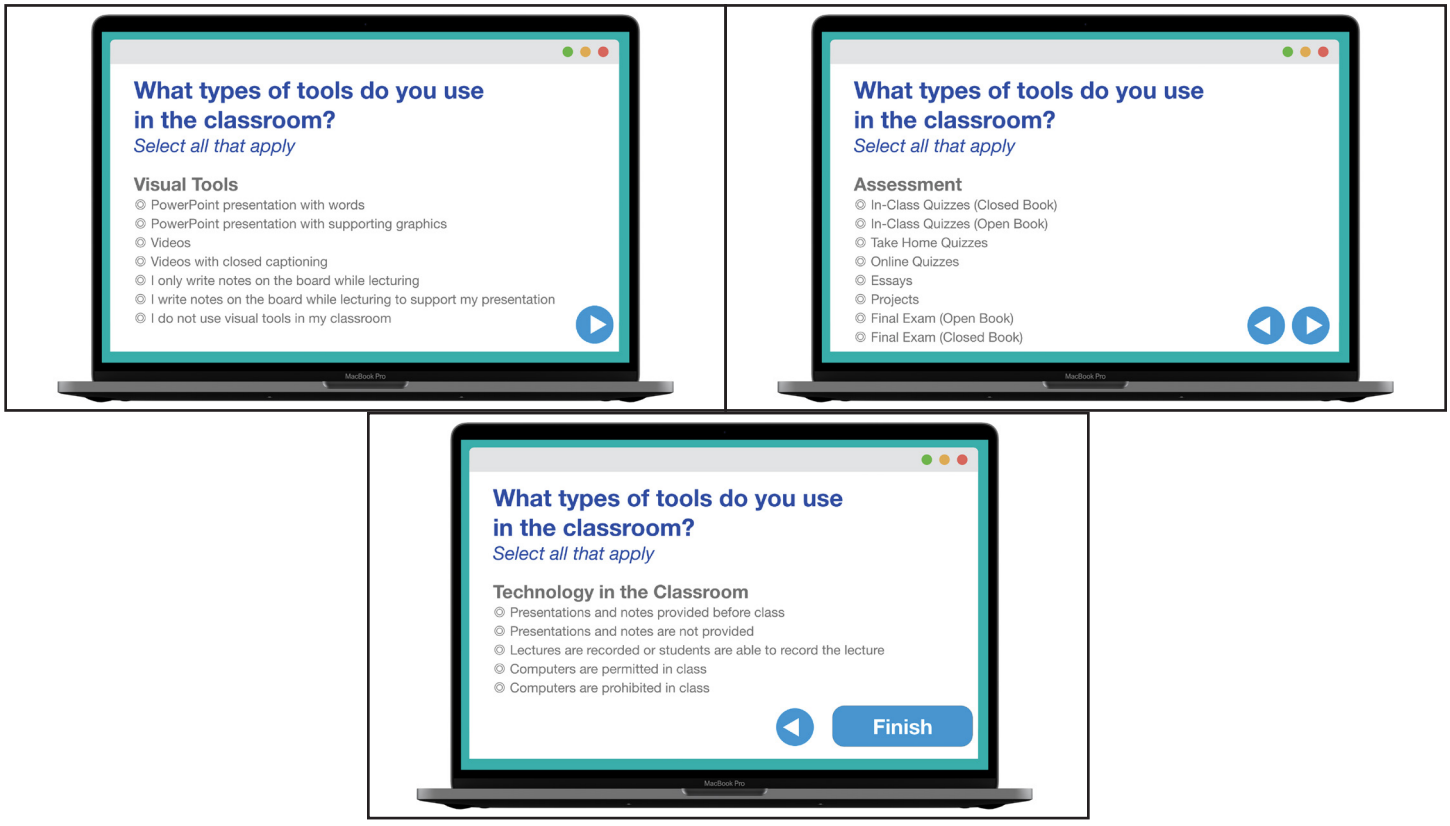
Key Takeaways:

- Instead of a poster, pamphlet or presentation, a website could be created
- Website should be interactive and designed specifically for professors looking to implement UDL in their classrooms
- Website could be a quiz where professors can select what types of teaching methods they use that then generates a list of recommendations
- Classmates liked this idea

6 LOW-FIDELITY PROTOTYPE

This prototype was created to represent what a website could look like. In the prototype, professors could select what tools they use in their classroom for teaching. When they click "Finish", it would provide them with recommendations for implementing Universal Design for Learning in their classrooms.

Interactive Questionnaire



Results

